CHAPTER 7

GCD, Bézout's and Gauss's theorems

Euclid's algorithm, coprime integers, Diophantine equations

Objectives. Find a gcd with Euclid's algorithm; obtain a Bézout pair; use Bézout's and Gauss's theorems; find an inverse modulo n; solve an equation $ax + by = c$ in integers.

Prerequisites. Chapter 6 (divisibility, Euclidean division, congruences); programming a loop in Python.

Fluency

Fluency · No calculator, five minutes.

1. Carry out the Euclidean division of 1,053 by 325.
2. Give the greatest common divisor of 84 and 126.
3. Do the numbers 15 and 28 have a common divisor other than 1?
4. Find two integers u and v such that $5u + 3v = 1$.
5. Solve $4x \equiv 0 \pmod 6$ for $x \in \{0, \dots, 5\}$.
6. If 7 divides $5k$, what can be conjectured about k?

Introductory activity

Activity 1 · Measuring one litre with two cans 35 min. We have a 9-litre can, a 7-litre can, both without markings, and a water supply. We may fill a can, empty it, or pour one into the other until one is empty or the other is full.

1. Show that one can get exactly 2 litres in the large can.
2. If we pour in u times 9 litres altogether and throw away v times 7 litres, the quantity left is $9u - 7v$. Find natural numbers u and v such that $9u - 7v = 1$. Deduce how to get 1 litre.
3. The cans are replaced by cans of 6 and 9 litres. Show that every quantity of the form $6u + 9v$, with u and v integers, is a multiple of 3. Can one get 1 litre?
4. With cans of a and b litres, which whole quantities does it seem possible to get? State a conjecture involving the greatest common divisor of a and b.

Observing is not proving: the properties conjectured here are established in the course.

Course notes

GCD and Euclid's algorithm

Definition · GCD. Let a and b be two integers, not both zero. The set of their positive common divisors is finite and contains 1; its largest element is called the *greatest common divisor* of a and b, written $\gcd(a,b)$. We say that a and b are *coprime* if $\gcd(a,b)=1$.

Property · Gcd and Euclidean division. Let a and b be two integers with $b\neq 0$. For every integer k, the common divisors of a and b are those of b and $a-kb$. In particular, if r is the remainder of the division of a by $b\geq 1$, then $\gcd(a,b)=\gcd(b,r)$.

Proof. If $d\mid a$ and $d\mid b$, then $d\mid a-kb$. Conversely, if $d\mid b$ and $d\mid a-kb$, then $d\mid (a-kb)+kb=a$. The two sets of common divisors are equal, so their largest elements are too. □

Method · Euclid's algorithm. Replace (a,b) by (b,r) as long as the remainder r is not zero. The remainders form a strictly decreasing sequence of natural numbers, so the algorithm stops. The gcd is the last nonzero remainder. Example:

$$1{,}053=3\times 325+78,\qquad 325=4\times 78+13,\qquad 78=6\times 13+0,$$

so $\gcd(1{,}053,325)=13$.

Bézout's theorem

Property · Writing the gcd. Let a and b be two integers, not both zero. There are integers u and v such that $au+bv=\gcd(a,b)$. Moreover, every common divisor of a and b divides $\gcd(a,b)$.

Proof. Let E be the set of positive integers of the form $au+bv$, with u, v integers. E is not empty (it contains a^2+b^2). Let $d=au_0+bv_0$ be its smallest element.

Divide a by d: $a=dq+r$ with $0\leq r<d$. Then $r=a-q(au_0+bv_0)=a(1-qu_0)+b(-qv_0)$ is of the form $au+bv$. If it were positive, it would belong to E while being smaller than d: impossible. So $r=0$ and $d\mid a$. Likewise $d\mid b$.

Finally every common divisor c of a and b divides $au_0+bv_0=d$, so $c\leq d$. Hence d is the greatest common divisor: $d=\gcd(a,b)$. □

Property · Bézout's theorem. Two integers a and b are coprime if and only if there are integers u and v such that $au+bv=1$.

Proof. If $\gcd(a,b)=1$, the previous theorem provides u and v. Conversely, if $au+bv=1$, every positive common divisor of a and b divides 1, hence equals 1. □

Method · Bézout pair: working back through Euclid's algorithm. With the previous example: $13=325-4\times 78=325-4(1{,}053-3\times 325)=13\times 325-4\times 1{,}053$. A suitable pair for $(1{,}053,325)$ is $(-4,13)$.

Key point · Extended Euclidean algorithm in Python. The following function returns (d,u,v) with $d=\gcd(a,b)=au+bv$, for natural numbers a and b not both zero.

```
def euclide_etendu(a, b):
    u0, v0, u1, v1 = 1, 0, 0, 1
    while b != 0:
        q, r = a // b, a % b
        a, b = b, r
```

```
        u0, u1 = u1, u0 - q * u1
        v0, v1 = v1, v0 - q * v1
    return a, u0, v0
```

Let a_0 and b_0 be the initial values. At each step, the variables `a` and `b` are equal to $a_0u_0 + b_0v_0$ and $a_0u_1 + b_0v_1$.

Gauss's theorem

Property · Gauss's theorem. Let a, b, c be three integers. If $a \mid bc$ and a and b are coprime, then $a \mid c$.

Proof. By Bézout's theorem, there are u and v such that $au + bv = 1$. Multiplying by c: $c = acu + bcv$. Now $a \mid acu$ and $a \mid bc$, so a divides $acu + bcv = c$. □

Property · Corollary. If a and b are coprime and both divide n, then ab divides n.

Proof. $n = ak$. Since $b \mid ak$ and $\gcd(a, b) = 1$, Gauss gives $b \mid k$: $k = bk'$ and $n = abk'$. □

Property · Inverse modulo n. Let $n \geq 2$ and a an integer. There is an integer u such that $au \equiv 1 \pmod{n}$ if and only if $\gcd(a, n) = 1$. Such a u is an *inverse of a modulo n*; it allows us to solve $ax \equiv b \pmod{n}$: $x \equiv ub \pmod{n}$.

Proof. $au \equiv 1 \pmod{n}$ is equivalent to the existence of an integer v such that $au + nv = 1$: this is Bézout's theorem. □

Method · Equation $ax + by = c$ in integers.

1. It has solutions if and only if $\gcd(a, b)$ divides c; we then divide by this gcd.
2. Find a particular solution (x_0, y_0).
3. By subtraction, $a(x - x_0) = -b(y - y_0)$; with a and b coprime and non-zero, Gauss gives $x - x_0 = bk$, then $y - y_0 = -ak$.
4. Check that the pairs $(x_0 + bk, y_0 - ak)$, $k \in \mathbb{Z}$, are indeed solutions.

Example: $7x + 5y = 1$ has the particular solution $(-2, 3)$; its solutions are the pairs $(-2 + 5k, 3 - 7k)$, $k \in \mathbb{Z}$.

Exercises

Exercise 7.1 · Euclid's algorithm. Find $\gcd(1{,}071, 462)$ with Euclid's algorithm, then simplify the fraction $\dfrac{462}{1{,}071}$.

Exercise 7.2 · Bézout pair. Find a pair of integers (u, v) such that $17u + 12v = 1$.

Exercise 7.3 · Coprime integers. Let n be a natural number. Prove that n and $n+1$ are coprime, then that $2n+1$ and $3n+1$ are coprime too.

Exercise 7.4 · A homogeneous equation. Find all pairs of integers (x, y) such that $15x = 21y$.

Exercise 7.5 · Exercise books and pens. A bookshop sells exercise books at 1,100 ariary and pens at 700 ariary. A customer spends exactly 10,000 ariary. How many exercise books and pens did they buy?

Exercise 7.6 · Inverse modulo 26.

1. Explain why 7 has an inverse modulo 26 and check that 15 is one.
2. Solve $7x + 3 \equiv 10 \pmod{26}$.

Exercise 7.7 · A gcd that depends on n. Let n be a natural number and $d = \gcd(n+5, 2n+3)$.

1. Prove that d divides 7.
2. Find the integers n for which $d = 7$.

Problem

Problem · The affine cipher. The letters A to Z are numbered from 0 to 25. To encrypt a letter with number x, compute the remainder y of $7x + 3$ in the division by 26, and replace the letter by the letter with number y.

1. Encrypt the word EXPERT.
2. Prove that $y \equiv 7x + 3 \pmod{26}$ is equivalent to $x \equiv 15y + 7 \pmod{26}$.
3. Decrypt the message TDNZZ.
4. Replace 7 by a in the key. Prove that if $\gcd(a, 26) \neq 1$, two different letters may be encrypted as the same letter. How many values of $a \in \{1, \ldots, 25\}$ give a cipher that can be decrypted?